6 **(a)** Show that the equation $\sin\left(x+\frac{1}{6}\pi\right)=\cos\left(x-\frac{1}{4}\pi\right)$ can be written in the form

$$\tan x=\frac{\sqrt{2}-1}{\sqrt{3}-\sqrt{2}}.$$ **[4]**

(b) Hence solve the equation $\sin\left(x+\frac{1}{6}\pi\right)=\cos\left(x-\frac{1}{4}\pi\right)$ for $0\leqslant x\leqslant 2\pi$. **[1]**